

Total No. of Questions :12]

SEAT No. :

P2196

[5058]-309

[Total No. of Pages : 6

**T.E. (Civil Engineering)
STRUCTURAL DESIGN -II
(2012 Course) (Semester-II)**

Time : 3 Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10 and Q.11 or Q.12.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Use of IS 456-2000 and non programmable calculator is allowed.*
- 5) *Mere reproduction from IS code as answer, will not be given full credit.*
- 6) *Assume suitable data, if necessary.*

Q1) a) Draw strain and stress distribution diagrams with all parameters for the design of RCC section of flexural member using LSM. **[3]**

b) Explain need of Under reinforced section in L.S.M. Design. **[3]**

OR

Q2) Explain the terms bond stress and development length. Calculate development length for 16mm diameter bar in compression and tension by both methods (WSM and LSM). Use M20 concrete and Fe 500 steel. **[6]**

Q3) Calculate the moment of resistance by LSM for flanged beam section detailed as below: **[8]**

- a) Width of rib = 230 mm
- b) Effective flange width = 1400 mm
- c) Thickness of flange = 125 mm
- d) Effective depth = 565 mm
- e) Tension steel = 2 - #20 through plus 2 - #16 curtail at midspan.
- f) Use M20 grade of concrete and Fe 415 grade of steel.

OR

P.T.O.

- Q4)** A rectangular beam section, 230 mm wide and effective depth 415 mm is reinforced with 3 bars of 20 mm diameter in the tensile zone and 2 bars of 16 mm in the compression zone. Determine moment of resistance of the section using WSM. Use M20 grade of concrete and Fe 415 grade of steel. [8]
- Q5)** Design a cantilever slab for effective span of 1.3 m subjected to floor finish of 2 kN/m² and live load 3 kN/m². Use Concrete of grade M20 and Fe 500 reinforcement. Draw details of reinforcement. Check for shear is not required. (Use LSM). [8]

OR

- Q6)** Design a simply supported slab for a room with clear inner size 3 m x 7 m. The slab is supported by beams of width 230 mm along all the edges. The slab is subjected to floor finish of 1 kN/m² and live load 4 kN/m². Use Concrete of grade M20 and Fe 500 reinforcement. Draw details of reinforcement. Check for shear is not required. (Use LSM). [8]
- Q7)** Continuous RC beam ABCD of rectangular section is simply supported at A and D and continuous over support B and C. Span AB = 4.5 m, BC = 6.5 m and CD = 5.5 m. The beam carries dead load of 20 kN/m (including its self weight) and live load of 16 kN/m. The beam supports 120 mm slab on both sides. Calculate design moment for span BC after 20% redistribution of moments by considering proper load case. Design span BC for flexure and shear. Draw the reinforcement details. Material-Concrete of grade M25, Fe 500 reinforcement. [16]

OR

- Q8)** Design a continuous beam ABCD for flexure only using IS Code coefficients. AB = BC = CD = 4.2 m. The beam supports 120 mm slab on both sides. The beam carries dead load of 18 kN/m (including its self-weight) and live load of 10 kN/m. Take material M20 and Fe 500. Show the reinforcement detail in longitudinal section and cross-section at continuous support and at mid span. [16]
- Q9)** A rectangular RC beam of span 6m, size 300 mm x 600 mm with effective cover 40 mm is subjected to following actions: [16]
- a) Factored BM = 90 kN.m
 - b) Factored SF = 60 kN
 - c) Factored Torsional Moment = 35 kN.m

Design the beam for flexure and shear using M 25 & Fe 500 grade materials.

OR

Q10) Design an axially loaded short column to carry a working load of 950 kN. The unsupported length of column is 3.5 m. The column is held in position and not restrained against the rotation at both ends. Also design the footing for this column only for flexure and one way shear. Take $SBC = 200 \text{ kN/m}^2$.

Material M 25 and Fe 500 used. Show detailed load and design calculations and reinforcement details in plan and sectional elevation. **[16]**

Q11) Design a bi-axial rectangular short column by limit state method with material M25 and Fe 415 to carry a working load of 900 kN. Working moment of 90 kN-m about major axis bisecting the depth of column and 35 kN-m about minor axis bisecting the width of column. The unsupported length of column about major and minor axis is 3.6 m and 3.2 m. The column is fixed at one end and hinged at the other. Show detailed design calculations and reinforcement details. **[16]**

OR

Q12) Design an uniaxial square short column by limit state method with material M25 and Fe 500 to carry ultimate load of 900 kN and working moment of 75 kN-m about major axis bisecting the depth of column. The unsupported length of column is 3.6m. The column is fixed at one end and hinged at the other. Also design the footing for this column only for flexure and punching shear. Take $SBC = 225 \text{ kN/m}^2$. Show detailed design calculations and reinforcement details in plan and sectional elevation. **[16]**

Chart 5 : Interaction Diagram for Combined Bending and Compression Rectangular Section-Equal Reinforcement on All Sides

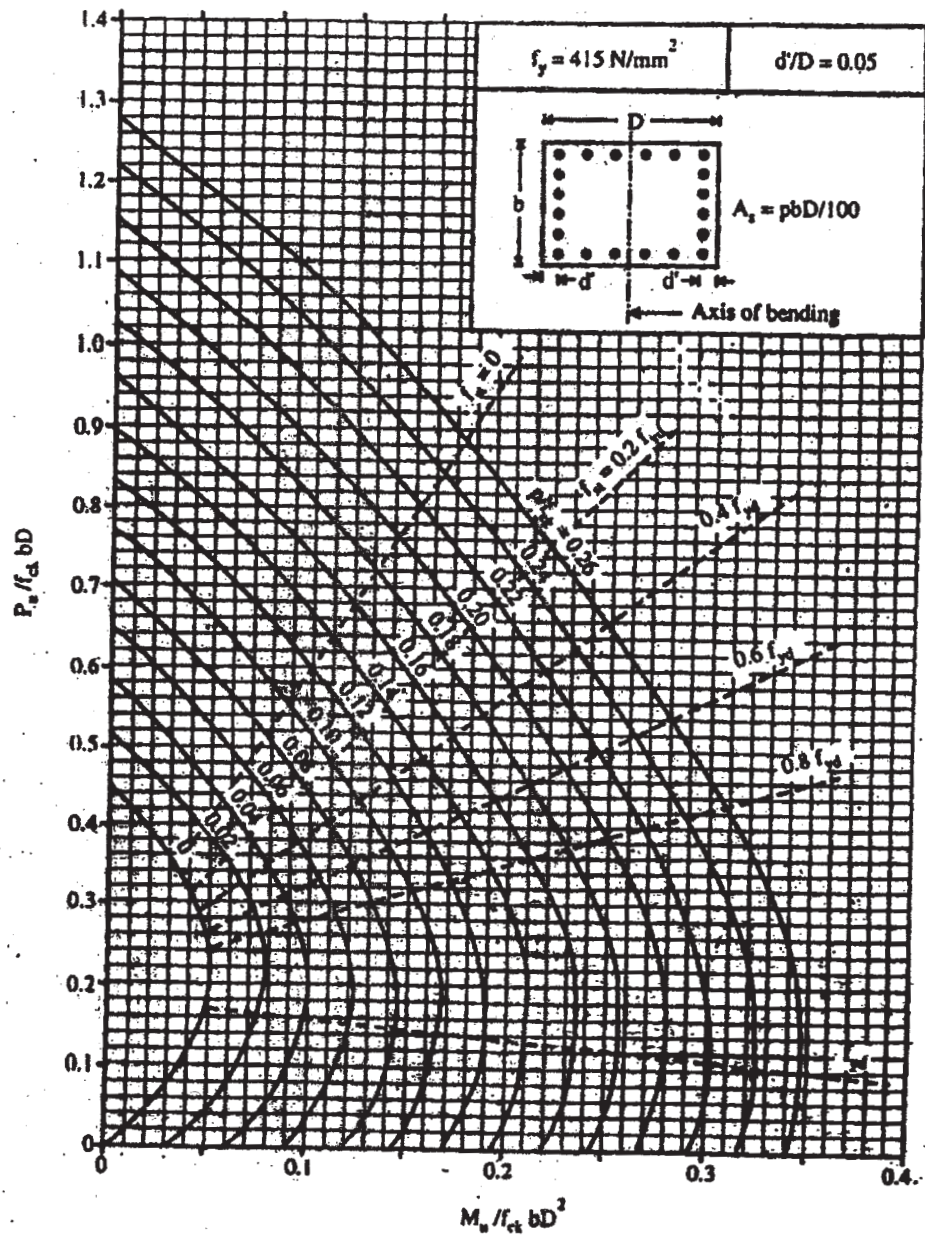


Chart 5.

Chart 6 : Interaction Diagram for Combined Bending and Compression Rectangular Section-Equal Reinforcement on All Sides

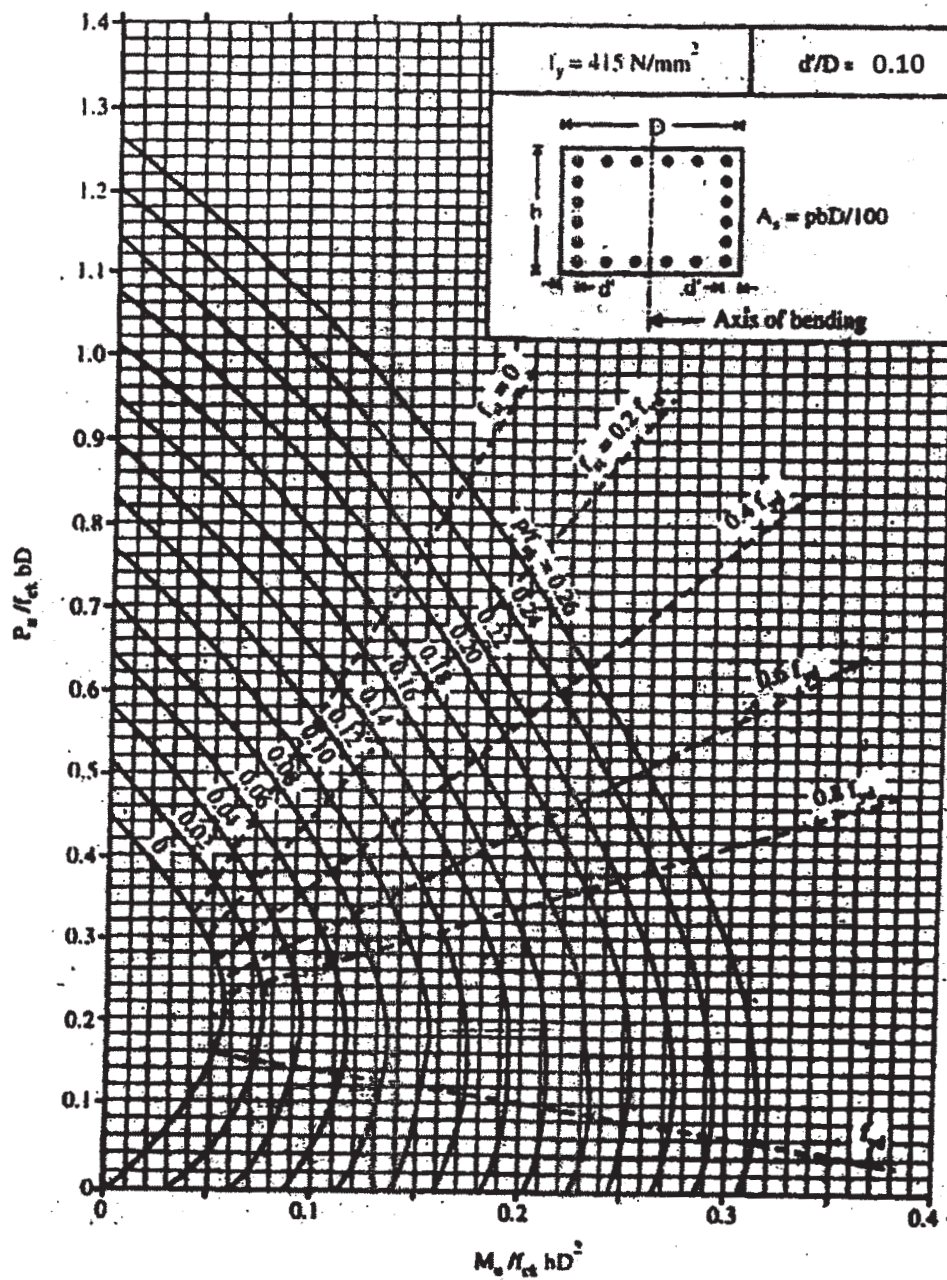
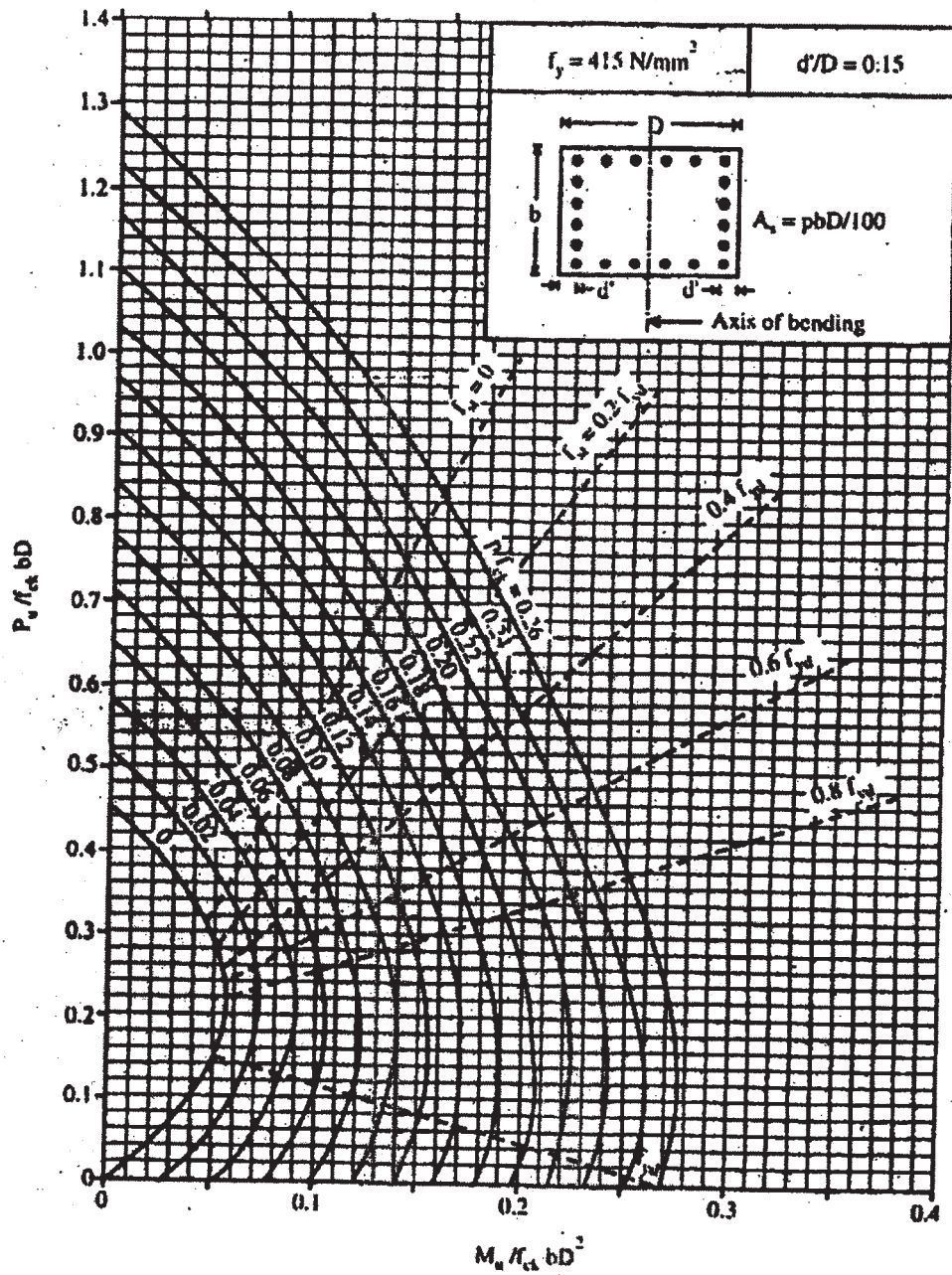


Chart 6

Chart 7 : Interaction Diagram for Combined Bending and Compression Rectangular Section-Equal Reinforcement on All Sides



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